

Solution Code - Engineering concepts - Algorithms_DataStructures - Inventory_Management_System

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: InventoryManager.java

```
1 | import java.util.HashMap;
2 |
3 | class Product {
4 |     String productId;
5 |     String productName;
6 |     int quantity;
7 |     double price;
8 |
9 |     public Product(String id, String name, int q, double p) {
10 |         this.productId = id;
11 |         this.productName = name;
12 |         this.quantity = q;
13 |         this.price = p;
14 |     }
15 | }
16 |
17 | public class InventoryManager {
18 |     private HashMap<String, Product> inventory = new HashMap<>();
19 |
20 |     public void add(Product p) { inventory.put(p.productId, p); }
21 |     public void update(String id, int quantity) { if (inventory.containsKey(id))
inventory.get(id).quantity = quantity; }
22 |     public void delete(String id) { inventory.remove(id); }
23 |
24 |     // Analysis: HashMap has O(1) average time complexity for add, update, delete operations.
25 | }
```